Server2 push delivery

Editable operator guide for S18b-2 and S18b-3 (#2703), 2026-10-16. PDF and sequence.

Capability and setup boundary

The AlertWorker library evaluates conditional registrations using server2's acknowledged S3 state. It is not wired into the production entrypoint in this slice. S20 must create one delivery owner, connect SERVICE_SERVER to server2's v000902 routes and provide the same S18b-1 Firebase account configuration. Do not start legacy and server2 senders together. There is no dry-run, mock or validation mode. Tests and the example inject synthetic peers.

Selection and outcomes

The scheduler runs at UTC minutes 7, 17, 35 and 50. Registrations must be explicitly enabled and inside their inclusive normal or overnight window. A weather or air push within the last six hours suppresses that registration. Current precipitation and current air-grade transitions can send; two successive wet shortest forecasts can trigger precipitation. Air forecasts are computed but discarded, matching legacy. Overseas observations older than 30 minutes are skipped.

The worker claims the cycle in S3 before preparation and stores alert state before FCM. It skips a registration changed by the user while weather was fetched. Each provider submission is attempted once. Unknown outcomes and rejected submissions are not retried after restart. An unregistered token disables all matching conditional registrations, without touching alarms. Diagnostics omit tokens, names, coordinates and notification text.

Outcome	Operator meaning
accepted	Local/provider API accepted the submission; this does not prove device delivery
rejected	Provider rejected it; do not resend this cycle
unknown	Outcome may have been delivered; do not resend
failed / state-not-committed	State could not be safely committed or the selected registration changed; nothing sent
not-needed	State was updated; no notification threshold transition

Local functional check

From server2/, run:

```
cargo run --locked --example conditional_alert_smoke
cargo test --locked --test delivery alerts::
```

The example creates one synthetic alert and local weather, OAuth and FCM HTTP peers. It fetches weather, durably updates the alert, submits once and restores the S3 state into a new service. It expects one weather GET, one local FCM submission and zero restart resends. It reads no .env, real credentials or real device tokens and makes zero external calls.

```
$ cargo run --locked --example conditional_alert_smoke

{
  "alert_journal_durable": true,
  "clock": "2026-10-16T00:07:00Z",
  "external_calls": 0,
  "local_fcm_submissions": 1,
  "outcome": "PASS",
  "restart_resends": 0,
  "scenario": "conditional-alert-and-restart",
  "weather_http_gets": 1
}
```

Rendered capture of actual local smoke stdout; synthetic peers only.

Deployment and failures

This PR does not deploy, attach a scheduler or send live notifications. Do not infer a live FCM credential check or native device receipt from this local example. A refused cycle claim or alert-state journal write prevents submission. A lost result object keeps the claim and must never be replayed as a send. Ordinary conditional evaluations preserve registration updatedAt; legacy's obsolete updateAt cleanup typo remains a no-op.

Selected verification contains source provenance, actual commands and limitations. Google documents the HTTP v1 send contract and FCM error classes; observed provider acceptance and real device delivery remain separate facts.

Warning feed (S18b-3 library)

This slice provides the severe-warning feed and one-shot worker; it does not start a live process. S20 installs WarningCollector::with_push_feed and a single WarningWorker owner alongside the existing alarm/conditional workers. Its region adapter must use server2's native /geocode/v000903/coord/ route at cutover.

The first persisted feed and the first worker read are silent. Afterwards a fresh issued/changed warning selects enabled alert registrations inside their time window and current S12 warning zone. Multiple city registrations for one package/device receive one attempt. Release, expiry, a concurrent user update or an unavailable claim stops submission. Rejected, unknown or cancelled work is not automatically retried; restarting cannot resume it. An independently issued later warning can send again. An unregistered token disables alerts only.

S3 objects: index/v2/push/warning-feed/head.json, events/<identity>.json, and index/v2/push/delivery/warning/<campaign>/<endpoint>/<UTC-date>/<source-minute> claim/results. Claims and diagnostics contain hashes/record IDs and status classes, not token, location, name or notification text. Keep these claims during recovery; do not delete one to force delivery. Region and immutable-event memory caches can be discarded; S3 remains authoritative. No warning local-disk capacity is required.

For local functional verification, the maintained example uses actual loopback S3, geocode and OAuth/FCM HTTP peers plus synthetic registrations/service-account data:

```
cd server2
cargo run --locked --example warning_push_smoke
```

It never loads .env or calls a real provider/device. Successful execution must report one local submission and zero restart resends. This does not prove actual EC2-role permissions, deployed startup wiring or device delivery. New warning paths already fit the checked-in IAM/bucket index/v2/* policy; no AWS mutation is needed.

After OAuth preparation, the final guard checks the latest S12 durable zone state and catches up registration journals from S3, including other-process changes. Disabled/changed registration, released warning, expired five-minute window or an unavailable/timed-out guard results in no POST. The delivery claim remains terminal.

```
$ cargo run --locked --example warning_push_smoke

{
  "clock": "2026-10-16T01:02:00Z",
  "external_calls": 0,
  "geocode_http_gets": 1,
  "local_fcm_submissions": 1,
  "outcome": "PASS",
  "restart_resends": 0,
  "s12_raw_catalog_http": true,
  "s3_http_feed_and_claims": true,
  "scenario": "warning-feed-and-restart"
}
```

Rendered capture of actual local smoke stdout; synthetic peers only.

Warning verification records the actual legacy oracle and isolated failure checks. The image is a rendered capture of actual stdout, with synthetic names and no device token or private credentials.